Intelligent Customer Support Triage & Analysis System

Combining Transformers, TF-IDF Search, and Sentiment Intelligence

This system focuses on automating ticket routing, entity extraction, and emotional prioritisation to streamline customer support operations.

Project Objectives: Transforming Raw Data into Actionable Insights

Our primary goal is to convert raw customer support tickets into actionable data through a sophisticated, multi-layered Natural Language Processing (NLP) pipeline.

1

Automated Classification

Efficiently route incoming support tickets to the correct functional department (e.g., Technical, Billing, Sales, etc.).

2

Entity Recognition

Accurately extract specific Brands (50+) and Products (50+) mentioned in tickets for immediate context and relevance.

3

Sentiment Intelligence

Detect and classify customer emotions, specifically "Angry" or "Sad," to trigger priority escalations for critical cases.

4

Efficiency Optimisation

Implement a "Search-First" semantic cache, enabling resolution of repetitive queries within milliseconds, drastically improving response times.

Data Insights: Understanding Our Customer Support Dataset

An in-depth Exploratory Data Analysis (EDA) was crucial to understanding the characteristics and challenges of our customer support records.

- **Volume:** We analysed a substantial dataset comprising 8,469 customer support records, providing a robust foundation for model training.
- **Balance:** The dataset exhibited a near-perfect distribution across 5 distinct categories (approximately 20% each), ensuring a balanced training environment and mitigating potential model bias.
- **Complexity:** Tickets featured variable text lengths, ranging from concise 5-word messages to detailed 500-word descriptions, necessitating optimised tokenisation strategies.
- **Observation:** Initial analysis revealed inherent noise within the dataset, highlighting the critical need for a comprehensive and robust data cleaning pipeline.

The Cleaning Pipeline: Mitigating Noise with Regex Logic

To ensure optimal model performance, a meticulous noise reduction strategy was implemented using custom Regex and text normalisation techniques.

System Phrase Filtering

Custom Regex patterns were developed to strip boilerplate text, such as automated disclaimers (e.g., "Note: seller is not responsible"), enhancing content relevance.

Text Normalisation

This involved converting all text to lowercase, removing URLs, and masking sensitive or dynamic placeholders (e.g., {product} becomes "product") to standardise input.

Deduplication

Sentence-level deduplication was applied to prevent repeated phrases from unduly influencing the model's weighting and understanding.

System Architecture: A Hybrid "Smart" Pipeline

Our intelligent system employs a three-layered hybrid approach, combining speed, depth, and precise knowledge retrieval for optimal performance.

1. The Fast Path: Semantic Search

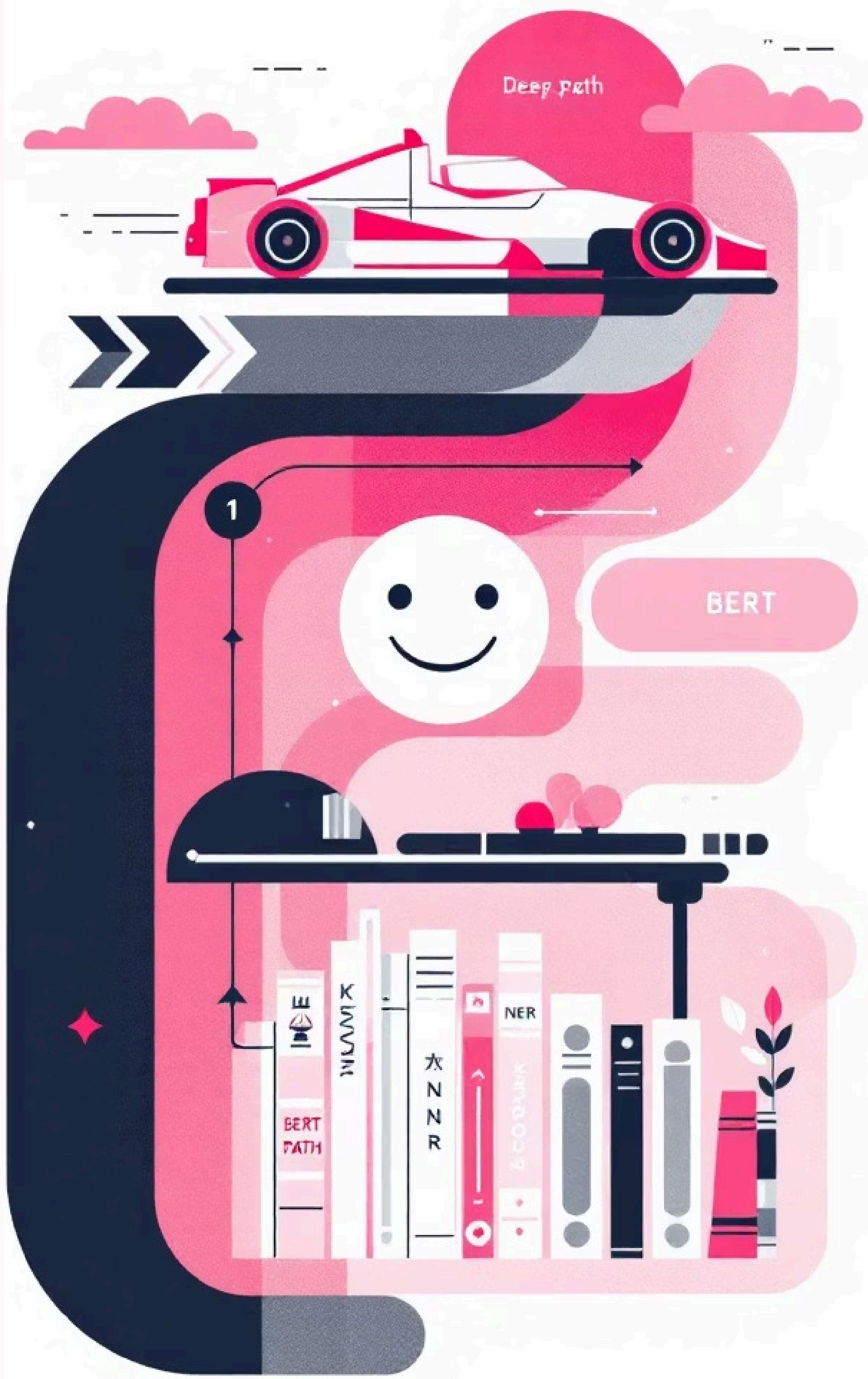
Utilises TF-IDF Vectoriser combined with Cosine Similarity to quickly identify historical matches with greater than 85% confidence, forming an efficient cache.

2. The Deep Path: BERT Inference

Leverages DistilBERT for two key functions: precise department classification and nuanced sentiment detection via the go_emotions framework.

3. The Knowledge Path: Rule-based NER

Employs a robust 100-keyword dictionary to achieve 100% precision in named entity recognition for known brands and products, ensuring accuracy.



Operational Business Logic: Automated Decision Making

The system integrates intelligent business rules to automate priority escalation and ensure critical issues are flagged for human review.

Priority Escalation

If the detected customer sentiment is identified as Anger, the ticket priority is automatically forced to HIGH, ensuring immediate attention.

Safety Net: Human Review Triggers

- If routing confidence falls below 0.50.
- If sentiment is determined to be Negative (Anger/Sadness), indicating a distressed customer.
- If no Product or Brand entities are successfully detected, suggesting missing context.

Sample Execution & Results: Real-world Scenario

Let's examine a typical customer query and observe how the system processes and prioritises it.

> "My Samsung Galaxy is not charging... very frustrating."

Department:	Technical Support (37.6% confidence)
Entities:	Brand: Samsung | Product: Phone
Sentiment:	Anger (99.6% confidence)
Result:	HIGH PRIORITY | HUMAN REVIEW TRIGGERED.

Critically, identical queries subsequently resolve in **<0.01s** due to efficient retrieval from the TF-IDF Cache, significantly boosting response efficiency.

Key Achievements & Future Scope: Towards Enhanced Intelligence

We have successfully developed a system that not only understands but also empathises with customer needs, while laying the groundwork for continuous improvement.

Key Achievements

- Successfully built a system demonstrating "Semantic Empathy," capable of understanding and prioritising user frustration effectively.
- Significantly reduced computational costs and enhanced query response times through the strategic implementation of TF-IDF caching.

Future Improvements

- Integration of advanced topic modelling to identify emerging issues and trends proactively.
- Development of a feedback loop for continuous model retraining and adaptation based on human agent corrections.
- Expansion of entity recognition to include more granular product features and customer-specific identifiers.
- Exploration of multilingual support to cater to a broader global customer base.

The Impact: Revolutionising Customer Support

This intelligent system transforms customer support from a reactive process into a proactive, empathetic, and highly efficient operation.

Enhanced Customer Satisfaction

Faster resolution times and empathetic handling of urgent queries lead to significantly happier customers.

Increased Operational Efficiency

Automation of routing and priority setting frees up agents to focus on complex issues, boosting overall productivity.

Deeper Business Insights

Granular data on customer issues, brands, and sentiment provides invaluable insights for product development and service improvements.

Thank You